Electric Power Systems Research 155 (2018) 80–92

Contents lists available at ScienceDirect

Electric Power Systems Research

journal homepage: www.elsevier.com/locate/epsr

European day-ahead electricity market coupling: Discussion, modeling, and case study

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ARTICLE INFO

Article history:
Received 17 March 2017
Received in revised form
28 September 2017
Accepted 1 October 2017
Available online 17 October 2017

Keywords:
European day-ahead electricity market
EUPHEMIA
MIQCP
Iterative procedure
Non-convexity
Public data

ABSTRACT

Currently, the integration of European Electricity Market (EEM) has led to a single European Day-Ahead Market (DAM) with multiple-areas considered as bidding zones. In the near future, the EEM will spread to the Intra-day and Balancing market. To operate the DAM, a market clearing tool (algorithm) has been developed by market operators. The development of this algorithm corresponds to three primary principles: (i) one single framework, (ii) robust operation, and (iii) individual accountability. However, this algorithm is not available to the research community. In this paper, the authors develop a complete European DAM model in General Algebraic Modelling System (GAMS), formulating it as a Mix Integer Quadratic Constraint Problem (MIQCP) and iterative procedure, to mitigate the non-convexity of electricity prices across Europe due to the "fill or kill" condition of block, complex and Prezzo Unico Nazionale (PUN) orders. Eventually, two case studies reflecting the current European DAM evaluated the model, aiming to confirm its robustness and reliability.

1. Introduction

In Europe, the spot market is operated by the Power Exchange (PX), while the Transmission System Operator (TSO) provides network boundary conditions, within which the system feasibility and security is guaranteed. Some PXs have split their market into several bidding zones based on national borders or network bottlenecks. For instance, the EPEX Spot has three bidding areas (France, Germany/Austria, and Switzerland) [1], while the GME (Italy) has more than 20 bidding areas [2].

The idea of a single energy market in Europe has been proposed in the Internal Energy Market (IEM) [3] project for the last two decades, to achieve three primary objectives: (i) affordable energy, (ii) competitive prices and (iii) environmental sustainability. In the first step of IEM, the Price Coupling of Regions project (PCR) has entirely coupled the DAMs of GME, EPEXSpot (only France, Germany and Austria in PCR), APX (UK and Netherlands), Belpex (Belgium), Nord Pool Spot (Scandinavian countries), OMIE (Spain and Portugal), and OTE (Czech Republic). The main PCR benefits [4,5] are (i) improving the market liquidity; (ii) guaranteeing the overall welfare, and (iii) implicit allocation management.

The PCR focuses on the development of a single market clearing algorithm based on the SESAM [6] and COSMOS [7], to calculate energy distribution and electricity price across Europe. The algorithm is named the Pan-European Hybrid Electricity Market Integration Algorithm (EUPHEMIA) [8], and its goals are to cover all specified conditions of every particular PX simultaneously and give solution within a reasonable time frame. Obviously, the combination of PXs could be risky if one looks at the coherence of their market rules. In details, one of the major problems is the non-convexity of electricity price due to the "fill-or-kill" condition of block, MIC and PUN orders. Moreover, the collaboration of the allocation methodologies (Available Transmission Capacity (ATC) and Flow-Based (FB) models), the false rejected block orders, and the decreasing of the fairness for market players because of the short computing time also put stress on EUPHEMIA.

1.1. Structure of the European day-ahead electricity market

1.1.1. Market orders

Paper [8] presents the structure and acceptance rule of all type of orders, including aggregated hourly, complex, block, and PUN orders [9]. Papers [10,11] show the main characteristics of the DAM in Europe. All the orders in the DAM are located in a Share Order Book (SOB) which is used as input data for the market clearing algorithm. In this subparagraph, theses orders and related acceptance rules are briefly described.

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https://doi.org/10.1016/j.epsr.2017.10.003
0378-7796/

Nomenclature

Acronyms

IEM Internal Energy Market
EEM European Electricity Market
DAM Day-Ahead Market
PCR Price Coupling of Regions project
EUPHEMIA Pan-European Hybrid Electricity Market Integration Algorithm
CWE Central Western Europe
GAMS General Algebraic Modelling System
PX Power Exchange
TSO Transmission System Operator
PUN Prezzo Unico Nazionale
LGC Load Gradient Condition
MIC Minimum Income Condition
SSC Schedule Stop Condition
ATC Available Transmission Capacity
FB Flow-Based
SOB Share Order Book
PAB Paradoxically Accepted Block order
PRB Paradoxically Rejected Block order
PAMIC Paradoxically Accepted MIC order
PRMIC Paradoxically Rejected MIC order
SW Social Welfare
RAM Remain Available Margin
PTDFs Power Transfer Distribution Factors
SWMP Social Welfare Maximization Problem
PAMIC-SP Paradoxically Accepted MIC Sub-Problem
PAB-SP Paradoxically Accepted Block Sub-Problem
PUN-SP PUN search Sub-Problem
PRMIC-SP Paradoxically Rejected MIC Sub-Problem
PRB-SP Paradoxically Rejected Block Sub-Problem

Sets

$t \in T$ set of dispatch periods in one day
$a \in A$ set of bidding areas

Sets of interconnectors

$cb \in CB$ set of all critical branches
$l \in Br_{fa.ta}$ set of interconnectors between areas $fa \in A$ and area $ta \in A$. The power flow is positively defined from fa to ta
$l \in Br_{fa.ta}^{DC}$ subset of DC interconnectors, including the ATC region ($Br_{fa.ta}^{ATC\_DC}$) and the FB region ($Br_{fa.ta}^{FB\_DC}$), $Br_{fa.ta}^{ATC+FB\_DC} = Br_{fa.ta}^{ATC\_DC} \cup Br_{fa.ta}^{FB\_DC}$, and the DC connections between ATC and FB region ($Br_{fa.ta}^{ATC.FB\_DC}$), $Br_{fa.ta}^{DC} = Br_{fa.ta}^{ATC+FB\_DC} \cup Br_{fa.ta}^{ATC.FB\_DC}$, $Br_{fa.ta}^{DC} \subseteq Br_{fa.ta}$
$l \in Br_{fa.ta}^{AC}$ subset of AC interconnectors, including the ATC region ($Br_{fa.ta}^{ATC\_AC}$) and the FB region ($Br_{fa.ta}^{FB\_AC}$), $Br_{fa.ta}^{ATC+FB\_AC} = Br_{fa.ta}^{ATC\_AC} \cup Br_{fa.ta}^{FB\_AC}$, and the AC connections between ATC and FB region ($Br_{fa.ta}^{ATC.FB\_AC}$), $Br_{fa.ta}^{AC} = Br_{fa.ta}^{ATC+FB\_AC} \cup Br_{fa.ta}^{ATC.FB\_AC}$, $Br_{fa.ta}^{AC} \subseteq Br_{fa.ta}$
$l \in Br_{fa.ta}^{ATC}$ subset of interconnectors in ATC region, where $Br_{fa.ta}^{ATC} = Br_{fa.ta}^{ATC\_AC} + Br_{fa.ta}^{ATC\_DC}$, $Br_{fa.ta}^{ATC} \subseteq Br_{fa.ta}$
$l \in Br_{fa.ta}^{FB}$ subset of interconnectors in FB region, where $Br_{fa.ta}^{FB} = Br_{fa.ta}^{FB\_AC} + Br_{fa.ta}^{FB\_DC}$, $Br_{fa.ta}^{FB} \subseteq Br_{fa.ta}$
$l \in ls$ set of interconnectors that are subject to the same constraint
$ls \in LS$ set of all ls
$l \in Br_{fa.ta}^{HR}$ subset of interconnectors that subject to the hourly ramping constraint, $Br_{fa.ta}^{HR} \subseteq Br_{fa.ta}$

Sets of orders

$s \in S_a$ set of hourly step orders submitted at bidding area a, including stepwise $ss \in SS_a$ and piecewise hourly order $sp \in SP_a$, where $S_a = SS_a \cup SP_a$
$ss \in SS_a^{pun}$ set of PUN orders, $SS_a^{pun} \subseteq SS_a$
$b \in B_a$ set of block orders submitted at bidding area a
$pb \in b$ set of small block orders which define block order b
$eg \in EG_a$ set of exclusive group orders including many regular block orders submitted at bidding area a
$lb \in LB_a$ set of link block orders submitted at bidding area a
$fh \in FH_a$ set of flexible hourly orders submitted at bidding area a
$p \in P_a$ set of producers submitted at bidding area a, it can be unit or portfolio
$p \in LGC_a^p$ set of producers p subject to LGC, where $LGC_a^p \subseteq P_a$
$p \in MIC_a^p$ set of producers p subject to MIC, where $MIC_a^p \subseteq P_a$
$p \in SSC_a^p$ set of producers p subject to SSC, where $SSC_a^p \subseteq P_a$

Parameters

P_{ss}^t, Q_{ss}^t price and quantity of step hourly order ss in period t, in €/MWh and MWh, respectively
$P0/1_{sp}^t, Q_{sp}^t$ price and quantity of piecewise hourly order sp in period t, in €/MWh and MWh, respectively
P_b, Q_{pb} price and quantity of small block order pb belong to block order b in period t, in €/MWh and MWh, respectively
P_{fh}^t, Q_{fh}^t price and quantity of flexible block order fh in period t, in €/MWh and MWh, respectively

The value of submitted quantity is positive for demand side and negative for supply side

$\underline{T_{pb}}, \bar{T_{pb}}$ starting (ending) dispatch period of small block orders pb
$\underline{U_{pb}^t}, U_{pb}^{\bar{t}}$ starting (ending) unit steps of small block orders pb. Here, its value receives 1 if $t \geq T_{pb}$ and receives 0 if $t < T_{pb}$, where $\forall t \in T$
$R_b^{\min}$ minimum acceptance ratio of block order b in p.u; where: $0 \leq R_b^{\min} \leq 1$ for profile block order and $R_b^{\min} = 1$ for regular block order
IM_b^{lb} incidence matrix relating the link block order to its parent. Thus, the km-th element of IM_b^{lb} is 1 if $k \in b$ block order is the parent of $m \in lb$ link block order, otherwise is null
$IG_p(DG_p)$ maximum increase (decrease) gradient of producer p in MWh, where $IG_p \in \Re^+$, $DG_p \in \Re^-$, $\forall p \in LGC_p$
$FT_p(VT_p)$ fixed term (Variable term) of the MIC of producer p in € (€/MWh), $\forall p \in MIC_p$
$Ymic_p$ binary parameter indicating if the producer p subjects to MIC is located in share order book or not (1 is located and 0 is not)
$Yssc_{ss}^t$ status of hourly stepwise orders of producer subject to SSC in period t; it is equal to 1 for the first hourly sub-order in the first three periods, and 0 for the rest
$Br_l^{t,\max/\min}$ limit capacity of inter-connector l in period t in MWh, where $Br_l^{t,\max} \in \Re^+$; $Br_l^{t,\min} \in \Re^-$
$Inp_{l/ls/a}$ initial power of the interconnector l, the line set ls, the area a in the last hour of previous day in MWh, respectively
Inp_a^d the net position of area a in previous day in MW

$R^{t,U/D}_{l/ls/a}$	hourly ramping up and down limits for the inter-connector l, the line set ls, the bidding area a in the period t in previous day in MWh/h, respectively, where $R^{t,U}_{l/ls/a} \in \Re^{+}; R^{t,D}_{l/ls/a} \in \Re^{-}$
$R^{U/D}_{a}$	hourly ramping up and down limit of total energy for area a in one day in MWh, where $R^{U}_{a} \in \Re^{+}; R^{t}_{a} \in \Re^{-}$
$PTDF^{t}_{cb,a}$	the PTDFs of area a on critical branch cb in period t, where a is in the Flow-based region
RAM^{t}_{cb}	the RAM on critical branch cb in period t in MW
taf_l	flow tariff coefficient of interconnector l, in €/MWh
c_l	penalty coefficient applied in the objective function to obtain a correct decomposition of power flows into positive and negative
$loss_l$	loss coefficient of inter-connector l
Δ^{t}	PUN imbalance in period t in €
$\Delta_{min/max}$	minimum and maximum PUN imbalance tolerance in €
$MCPint^{t}_{a}$	market clearing price of day-ahead electricity market of one day before in €/MWh

Continuous variables

$x^{t}_{ss/sp}$	clearing acceptance ratio of hourly stepwise ss and piecewise sp order in dispatch period t in p.u.
x_b	clearing acceptance ratio of block order b in p.u.
x^{t}_{fh}	clearing acceptance ratio of flexible hourly order fh in period t in p.u.
f^{t}_{l}	power flow on inter-connector l in period t
$f^{t,+}_{l}, f^{t,-}_{l}$	positive and negative value of power flow on inter-connector l in period t, where $f^{t,+}_{l} \in \Re^{+}, f^{t,-}_{l} \in \Re^{-}$
P^{t}_{a}	net position of bidding area a in period t
MCP^{t}_{a}	Market Clearing Price of bidding area a in period t
PUN^{t}	Prezzo Unico Nazionale (National Uniform Price) in period t

Binary variables

Y_b	clearing status of profile and regular block order b
Y^{t}_{fh}	clearing status of flexible hourly order fh

Functions

c^{t}_{ss}	cost function of step hourly order ss in dispatch period t, in €/h
c^{t}_{sp}	cost function of piecewise hourly order sp in dispatch period t, in €/h
c_b	cost function of block order b in dispatch period t, in €
c^{t}_{fh}	cost function of flexible hourly order fh in dispatch period t, in €/h
$f^{t,in/out}_{l}$	power flow function on interconnector l, in/out of bidding area a in dispatch period t, in MWh

Complex orders. A complex order can be subjected to a *MIC* and/or a *LGC*, with or without *SSC*. This order is used in OMIE. Here, the MIC means that the revenue collected in all periods (Acquired Revenue – AR_p) must be higher than its Required Revenue – RR_p ($AR_p \geq RR_p$) for the order to be accepted:

$$AR_p = \sum_{a \in A}\sum_{t \in T}\left(MCP^{t}_{a} * \left(\sum_{ss \in p}|Q^{t}_{ss} * x^{t}_{ss}|\right)\right), \quad \forall p \in MIC^{p}_{a} \tag{1}$$

$$RR_p = FT_p + VT_p * \left(\sum_{t \in T}\sum_{ss \in p}|Q^{t}_{ss} * x^{t}_{ss}|\right), \quad \forall p \in MIC^{p}_{a} \tag{2}$$

where FT_p and VT_p are two parameters which are defined in the complex order.

Block orders. The simplest and most frequently used block orders are the *regular block orders* which is accepted or rejected based on submitted price. In the complex case, block orders can be defined as the *profiled block order, linked block orders, exclusive group, and flexible hourly orders*. The acceptance rule of profile block order is based on the block volume weighted average market clearing price for the periods for which the orders were accepted, namely Wap_b in (3) [4]. The supply (demand) block b is accepted if Wap_b is greater (smaller) or equal to the submitted price P_b:

$$Wap_b = \frac{\sum_{t \in T}\sum_{pb \in b} x_b * Q_b * (\underline{U^{t}_{pb}} - \bar{U^{t}_{pb}}) * MCP^{t}_{a}}{\sum_{t \in T}\sum_{pb \in b} x_b * Q_b * (\underline{U^{t}_{pb}} - \bar{U^{t}_{pb}})}, \quad \forall b \in B_a, a \in A \tag{3}$$

The link block orders is a set of block orders that have together a linked execution constraint. The relationship between "child block" and "parent block" is described in [8,12]. In particular, the parent block order is accepted even if the acceptance rule is violated if the child's SW^{c}_{b} can compensate the loss of welfare of the parent SW^{p}_{b} (4):

$$x^{p}_{b} > 0 \Rightarrow \sum_{b \in lb}[SW^{p}_{b} - SW^{c}_{b}] > 0; \ \forall lb \in LB_a \tag{4}$$

PUN orders. They are a particular type of demand stepwise orders, and they are cleared at PUN^{t} rather than at MCP^{t}_{a}. The PUN^{t} is equal to the average of the prices of the geographical areas, weighted according to the quantities purchased in these zones.

About the block and the MIC orders, there are two possible cases that can generate non-convexity due to the "fill or kill" and MIC condition. In the first case, the bid is accepted even if the acceptance rule is violated. This situation is called Paradoxically Accepted Block order (PAB) or Paradoxically Accepted MIC order (PAMIC). Secondly, the bid is rejected but the acceptance rule is satisfied, this is called Paradoxically Rejected Block order (PRB) or Paradoxically Rejected MIC order (PRMIC). The block orders belonging to link block orders are treated in a different way. Here, the total welfare of parent and child will classify it as PABs/PRBs or not.

1.1.2. Congestion management

In Europe, PXs incorporate congestion management in DAMs through two standard approaches [13]: the ATC and the FB methodology.

- *The ATC model* has been used by the most PXs in Europe and considers that there is an interconnector between two areas when these areas share a border crossed by at least one physical link. Especially, the power will not flow according to the Kirchhoff's laws but follow the commercial path defined from the low price area to the high price area.
- *The Flow-based model* was developed by both Central Western Europe (CWE) and Central Eastern Europe (CEE). While the development is standstill in CEE, it has been activated in CWE on the 21st May, 2015 [14]. Mathematically, the FB model is a set of linear constraints on the net exports of the bidding areas approximating the physical flows on the critical branches. For this, the PTDFs are used as the constant coefficient of the linear constraints [14–16]. Nevertheless, the application of FB model has contained an issue in the clearing of commercial congestion, namely non-intuitive exchanges [8], in which the resulting flow goes from

high price zone to low price zone. Even if the Social Welfare (SW) is maximized, this result may be unacceptable for market designers and participants because they perceive it as unfair.

1.2. Literature review

In recent years, several papers have been published about the European DAM model, with different approaches to find the optimal solution, focusing on two main issues of the model: (i) non-convexity and (ii) coverage of the market rules, including the acceptance rules and congestion allocation.

First, the non-convexity is due to the presence of binary variables modeling "fill-or-kill" condition and due to the constraints introduced by the MIC and PUN orders (clearing prices are decision variables). Several interesting approaches have been proposed to mitigate the non-convexity in [17–25] with different optimization methods to find the feasible solution: Mixed Integer Programming (MIP) [17]; Linear Problem (LP) [18]; Mix Integer Linear Problem (MILP) [20–24,26–28] and Mixed Complementary Problem (MCP) [29–31].

In [17], the proposed methodology is based on MIP; here, a primal problem is solved firstly, then the integer variables are fixed to their optimal value to obtain a convex program. Article [18] proposed a semi-Lagrangian approach and formulate the problem as LP to find the optimal price. The iterative bid cut heuristic (adding a constraint to cut off the infeasible bid selection) and an exact algorithm (adding a positive continuous slack variable for every executed block; instead of minimizing the prices, the model minimizes the sum of these variables) are presented and compared in [19]. Obviously, the iterative bid cut heuristic is faster but produces less fairness than the exact algorithm; because it is too fast, thus it can introduce PRBs. In [20], the authors approved a MIP-based, primal-dual formulation of the European DAM which deals explicitly with the handling of the PRBs. Here, the authors do not use auxiliary continuous and discrete variables and proposed a strengthened Bender cuts method for efficiently handling the branch-and-bound process of the MIP solver. The procedure performs better than [19] in terms of computation time. In general, papers [17–20] are limited to using only regular block orders. An iterative process to clear PABs and PAMICs is presented and applied in [21]. However, this approach can give a suboptimal solution.

In a different direction, papers [22–24] propose a new price scheme and formulated an MILP to find the solution. The uplift is introduced in [22,23] to find the Walrasian equilibrium, while paper [24] is avoiding the uplift by adding a constraint that ensures participants achieve positive profit. However, the uplift clearing price models are a different market operating approach with respect to the official one.

Another direction, paper [25] uses the shadow price to present the hidden cost of ramping limits which can be used to find the optimal result, but the proposed model is applied to only the LGC.

Other papers are focused on developing the full algorithm [26–31]. This makes difficult dealing with MIC and PUN orders which clearing conditions depend on the market clearing price. Generally, the DAM model is solved iteratively and the MIC and PUN orders clearing conditions are checked and applied at the end of each iteration. For this purpose, the DAM model is formulated as a MILP [26–28], or as a MCP [29–31].

With MILP, one of the significant contributions was introduced by Biskas, P. N., et al. [26,27], taking into account only the block, hourly single step orders, and ATC model. Although this model is interesting for researching, it is not suitable for real-life as it does not cover all the complex aspects of the PCR, later introduced in 2014 such as complex and PUN orders, FB model or the non-convexity issue of either PABs or PRBs. In 2016, the authors enhanced the algorithm to collaborate clearing conditions of the block and complex orders [28] but lack of PUN orders, FB model and enhancing fairness of either PRBs or PRMICs. The MIC orders are iteratively removed from the SOB using a heuristic process where parameters *X* and *Y* define the threshold and chance of acceptance, respectively, in order to speed up the clearing process and reduce PRMICs. However, parameters *X* and *Y* can not warranty the fairness for producers which are subjected to MIC because it potentially introduces some PRMICs that leads to a suboptimal solution.

On the other hand, paper [29] focuses on implementing block orders by MCP. Following the success of [29], the article [30] introduces the PUN search Sub-Problem (PUN-SP). Here an iterative procedure is implemented to find the solution: (i) an MILP is solved in an inner iteration to handle block and MIC orders while (ii) the PUN-SP is formulated as an MCP and solved in an outer loop.

Papers [26–28,30] do not focuses on the allocation management as they implement only the simpler ATC model. However, in [31] the same authors propose a model to integrate ATC model and the FB model adopting an MCP approach. The proposed model includes two main steps: (i) maximizing social welfare, (ii) maximizing the intuitive exchanges considering the limitation of critical branches. The congestion management model adopted in this paper does not consider a FB region power flow condition necessary to represent the biding areas connected by both ATC and FB constrained interconnectors: the sum of imports and/or exports inside the FB region needs to be null. This condition is neccessary when the ATC and FB regions are neatly separated, like it happens in the real market, and, hence, not all the critical branches of the network are represented by FB constraints. Moreover, [31] does not consider the energy losses on the DC interconnectors, which requires the decomposition of the power flow into two complementary components (positive and negative flows) raising the complexity of the model.

Last but not least, [32] compares between two mathematic models (LP and MCP), in term of computation time. Here the authors show clearly that the MCP model is computationally demanding since the computation time increases significantly with the increase of the number of orders, while for the LP, it remains almost constant.

1.3. Motivations, contributions, and structure of the paper

According to the well-known of the authors, all studies [17–31] have focused on various particular aspects of the European DAM [8]. Nevertheless, EUPHEMIA has been operated officially at the European level; but it still needs to improve notably to cope with the expanding of the European DAMs and the increasing number and complexities of the products since 2014 from EUPHEMIA 6.0 to EUPHEMIA 9.3 [10]. Thus, it has become an interesting topic, but one of the limitations for researchers is the lack of a full mathematical model as the official model is not available to the public, but only some general aspects are presented in the available documentation. Therefore, the aim of this article is to develop a unified European DAMs algorithm which complies with the following paramount principles:

1. *One optimizing methodology:* Formulation of a complete market model as a MIQCP where all market rules are represented. The MIQCP formulation not only provides similar computation characteristics as with MILP but also can accommodate the piecewise orders present in the CWE which does not appear in papers [21,26–31].
2. *Complete market rules integration:* dealing with MIC/PUN orders and clearing of PABs and PAMICs through the iterative process. Not like [30], the PUN-SP of this paper contains only a linear constraint, while the PUN-SP of the paper [30] contains the dual/complementarity conditions. Moreover, a realistic alloca-

tion management procedure which combines all the aspects of the ATC and FB models is introduced (the hybrid model). Here, the authors do not separate the network into synchronous areas [26–28,31] and consider the FB region power flow condition and the presence of the energy losses on the interconnectors [30]. Therefore, the proposed model is close to the reality.
3. *Optimized efficiency:* Optimized the methods which prevent PAMICs, search PUN price and filter for PRBs in comparison to EUPHEMIA [8] to obtain optimized results.
4. *Fairness:* To overcome the inherent disadvantages of the iterative process, the PRBs and PRMICs are re-executed at the end of the iteration. This procedure mitigates the PRBs and PRMICs of [21,28,30] which are using an iterative process to clear PABs and PAMICs enhancing thus the fairness for participants.
5. *Robustness:* Propose a robust model validated using realistic data and large numerical test data.

The organization of the paper is the following: Section 2 provides the mathematical formulations of the model, while Section 3 shows the result to ensure that the model is robustness and authentic. Eventually, Section 4 formulates the conclusion.

2. European day-ahead electricity market model

Generally, the DAM electricity market solution is found by solving an optimization problem, namely Social Welfare Maximization Problem (SWMP). The MIC, block orders and PUN balance conditions defined in Section 1.1.1 can not be guaranteed by the constraints of the problem as they directly depend on the market clearing price. In this paper, the market clearing prices are obtained from the Lagrange multipliers of the power balancing equations in the SWMP by the conventional methodology in [17] (dealing with binary variables), therefore it can only be known after the convergence is found. To mitigate this issue, various modules, further named sub-problems, have been defined and solved iteratively to adjust the settings of the SWMP.

2.1. Social welfare maximization problem

The aim of SWMP is to maximize the SW and remove non-intuitive flow through the intuitive patch.

2.1.1. Objective function

$$\begin{aligned}\text{Min}\Bigg[&-\sum_{a\in A}\Bigg[\sum_{b\in B_a}c_b\\&+\sum_{t\in T}\left(\sum_{ss\in SS_a}c_{ss}^t+\sum_{sp\in SP_a}c_{sp}^t+\sum_{fh\in FH_a}c_{fh}^t\right)\Bigg]\\&+\sum_{l\in Br_{fa.ta}}\sum_{t\in T}(taf_l+c_l)*(f_l^{t,+}-f_l^{t,-})\end{aligned} \tag{5}$$

In traditional day-ahead electricity market, the objective function is maximizing the SW. However, in this model, the objective function (5) is minimized. Therefore, it comprises the total cost of supply minus the total cost of demand plus the sum of cost of flow tariff and coefficient c_l. Appendix A describes in details the cost functions of hourly stepwise orders (A.1), hourly piecewise orders (A.2), block orders (A.3) and flexible hourly orders (A.4). As mentioned in the nomenclature, the submitted quantities signs are distinguished by their side; thus, the costs of demand side are actually positive, while the costs of supplier side are negative and, thus, the sum of the two gives the social welfare. Moreover, the penalty term c_l is considered in order to correctly decompose the flows in the FB region and DC interconnectors into positive $f_l^{t,+}$ and negative $f_l^{t,-}$. This means that when $f_l^{t,+}$ is non-null, $f_l^{t,-}$ needs to be null and vice-versa. Therefore, c_l for DC interconnectors and for FB interconnectors region were set to 0.001 while for the remaining interconnectors, c_l are null. This parameter was manually set such that the associated penalty term does not have a negative impact on the solution of the market, i.e. such that minimizing the flow of interconnectors instead of maximizing the social welfare is avoided.

Market order constraints

$$Y_b*R_b^{\min}\le x_b\le Y_b,\ \forall b\in B_a \tag{6}$$

$$x_{fh}^t-Y_{fh}^t=0,\ \forall fh\in FH_a,t\in T \tag{7}$$

$$\sum_{b\in eg}x_b\le 1,\ \forall eg\in EG_a \tag{8}$$

$$x_{lb}-\sum_{b\in B_a}IM_b^{lb}*x_b\le 0,\ \forall lb\in LB_a \tag{9}$$

$$\begin{aligned}&\sum_{ss\in p}(x_{ss}^t*Q_{ss}^t*\delta^{t-1}-x_{ss}^{t-1}*Q_{ss}^{t-1})+DG_p^t\le 0,\\&\forall p\in LGC_a^p,t\in T\end{aligned} \tag{10}$$

$$\begin{aligned}&\sum_{ss\in p}(x_{ss}^t*Q_{ss}^t*\delta^{t-1}-x_{ss}^{t-1}*Q_{ss}^{t-1})+IG_p^t\ge 0,\\&\forall p\in LGC_a^p,t\in T\end{aligned} \tag{11}$$

$$\begin{aligned}&\sum_{ss\in PUN_{eq}^t}(Q_{ss}^t*x_{ss}^t)-\sum_{ss\in PUN_{eq}^t}(Q_{ss}^t*T_{pun}^t)=0,\\&\forall t\in T\end{aligned} \tag{12}$$

$$\begin{aligned}&x_{ss}^t\le\max(Ymic_p,\quad Yssc_{ss}^t),\\&\forall ss\in p,p\in SSC_a^p,t\in T\end{aligned} \tag{13}$$

Eqs. (6)–(11) model the clearing rules of regular, profile, link block, flexible hourly order and LGC as in [28]. In (10) and (11), the first hour should not be constrained [8]; hence a parameter δ^t is used, and equal to 1 ($\forall t\in T$) and 0 ($\forall t\notin T$). Eq. (12) optimizes the accepted quantity of step orders which belong to set PUN_{eq}^t in term of total accepted quantity that is defined by T_{pun}^t, a parameter set by the PUN sub-problem. Finally, constraint (13) is used to decide which MIC orders are considered by the SWMP at a certain iteration. This is done by the MIC sub-problem procedure by acting on the binary parameter $Ymic_p$. Moreover, the SSC market rule [8] is considered through the use of the constant parameter $Yssc_{ss}^t$.

2.1.2. Power balance and network constraints

$$\begin{aligned}&\sum_{b\in B_a}\sum_{pb\in b}x_b*Q_{pb}*(\underline{U_{pb}^t}-\bar{U}_{pb}^t)+\sum_{s\in S_a}Q_s^t*x_s^t+\\&quad\sum_{fh\in FH_a}x_{fh}^t*Q_{fh}^t+P_a^t=0,\forall a\in A,t\in T\end{aligned} \tag{14}$$

The equality constraint (14) defines the net position for each bidding area a at period t. The negative value of net position means that the area is importing energy, whereas the positive value says that it is exporting energy. Parameters $\underline{U_{pb}^t}$, $\bar{U}_{pb}^t$ are used to enforce the acceptance ratio of block orders, i.e. x_b, only for the submitted intervals [26,27]. Here, the unit steps are equal to 1 for submitted intervals. Especially, MCP_a^t is given by the Lagrange multiplier of the equality constraint (14).

$$\sum_{l \in Br_{fa.a}^{ATC.FB\_DC}} f_l^{t,in} - \sum_{l \in Br_{a.ta}^{ATC.FB\_DC}} f_l^{t,out} + \sum_{l \in Br_{fa.a}^{ATC.FB\_AC}} f_l^t - \sum_{l \in Br_{a.ta}^{ATC.FB\_AC}} f_l^t + P_a^t - P_a^{t,tem} = 0, \ \forall a \in A, t \in T \tag{15}$$

$$\sum_{l \in Br_{fa.a}^{ATC+FB\_DC}} f_l^{t,in} - \sum_{l \in Br_{a.ta}^{ATC+FB\_DC}} f_l^{t,out} \sum_{l \in Br_{fa.a}^{ATC+FB\_AC}} f_l^t - \sum_{l \in Br_{a.ta}^{ATC+FB\_AC}} f_l^t + P_a^{t,tem} = 0, \quad \forall a \in A, t \in T \tag{16}$$

$$f_l^{t,+} \le Br_l^{t,\max}, \ \forall l \in Br_{fa.ta}, t \in T \tag{17}$$

$$Br_l^{t,\min} \le f_l^{t,-}, \quad \forall l \in Br_{fa.ta}, t \in T \tag{18}$$

$$f_l^t = f_l^{t,-} + f_l^{t,+}, \ \forall l \in Br_{fa.ta}, t \in T \tag{19}$$

In FB model, the sum of the exported or imported energy inside the FB region must be equal to zero [14]. Thus, the energy which only flows inside the FB region, denoted $P_a^{t,tem}$ and named "temporary net position" in (15), is defined. This equation defines the hybrid model as it represents the relation between FB and ATC models from the trading energy point of view. Eq. (16) expresses power balance at each bidding area in period t. The $f_l^{t,+}$ and $f_l^{t,-}$ in (17)–(19) are used only for $Br_{fa.ta}^{DC}$ and FB model as in [28].

In Europe, PXs model the losses of $Br_{fa.ta}^{DC}$ as percentage of power flow, defined by $loss_l$. Therefore, the power flow is described by two different functions which are $f_l^{t,in}$ (B.1) and $f_l^{t,out}$ (B.2) in the B.

The capacity constraints of ATC and FB models are described by (20) and (21), respectively:

$$Br_l^{t,\min} \le f_l^t \le Br_l^{t,\max}, \forall t \in T, l \in Br_{fa.ta}^{ATC} \tag{20}$$

$$\left| \sum_{a \in FB} PTDF_{cb,a}^t * P_a^{t,tem} \right| \le RAM_{cb}^t, \ \forall cb \in (CB \backslash CBA), t \in T \tag{21}$$

In the ATC model, the capacity is guaranteed by $Br_l^{t,\min}$ and $Br_l^{t,\max}$. Meanwhile, the FB model includes a set of linear constraints, defined by parameters RAM_{cb}^t and $PTDF_{cb,a}^t$, published by TSOs. Obviously, the initial set of CBA is an empty set and will be updated after each loop of the "intuitive patch", aiming to clear non-intuitive problem in the FB region.

$$f_l^t - f_l^{t-1} - Inp_l * (1 - \delta^{t-1}) - R_l^{t,U} \le 0, \quad \forall l \in BrHR_{fa.ta}, t \in T \tag{22}$$

$$f_l^{t-1} - f_l^t + Inp_l * (1 - \delta^{t-1}) + R_l^{t,D} \le 0, \quad \forall l \in BrHR_{fa.ta}, t \in T \tag{23}$$

$$\sum_{l \in ls} [f_l^t - f_l^{t-1} - Inp_{ls} * (1 - \delta^{t-1})] - R_{ls}^{t,U} \le 0, \quad \forall ls \in LS, t \in T \tag{24}$$

$$\sum_{l \in ls} [f_l^{t-1} - f_l^t + Inp_{ls} * (1 - \delta^{t-1})] + R_{ls}^{t,D} \le 0, \quad \forall ls \in LS, t \in T \tag{25}$$

$$P_a^t - P_a^{t-1} - Inp_a * (1 - \delta^{t-1}) - R_a^{t,U} \le 0, \quad \forall a \in A, t \in T \tag{26}$$

$$P_a^{t-1} - P_a^t + Inp_a * (1 - \delta^{t-1}) + R_a^{t,D} \le 0, \quad \forall a \in A, t \in T \tag{27}$$

$$\sum_{t \in T} P_a^t - Inp_a^d - R_a^D \ge 0, \forall a \in A \tag{28}$$

$$\sum_{t \in T} P_a^t - Inp_a^d - R_a^U \le 0, \forall a \in A \tag{29}$$

Last but not least, the network constraints (22)–(29) model various limitations of power flow on interconnectors and bidding areas. They include hourly energy ramping up and down on interconnectors (22) and (23), on group of interconnectors (24) and (25), on bidding areas (26) and (27) and daily ramping up and down of bidding areas (28) and (29). These inequality constraints guarantee that the energy can not vary suddenly and significantly between two consecutive periods to guarantee the security of network and generators. Notably, the model computes the power of the last period of previous day, denoted Inp_l, Inp_{ls}, Inp_a and the total dispatching energy of previous day Inp_a^d.

2.1.3. Intuitive patch

An iterative process is implemented based on the fundamental idea which is published in [14] to overcome the non-intuitive problem. The non-intuitive issue can be viewed as a negative welfare of each interconnector and expressed as in (30). Subsequently, the set of the active critical branches CBA will be updated with the branches with negative welfare.

$$W_l^t = f_l^t * (MCP_{ta}^t - MCP_{fa}^t), \ \forall l \in Br_{fa,ta}^{FB}, t \in T \tag{30}$$

The price differences between two bidding areas are proportionate to the difference of PTDFs [14]. Thus, the new inequality constraint (31) substitutes (21), and it is applied $\forall cb \in CBA$. This constraint will minimize the negative SW of the branches, preventing adverse flows:

$$\sum_{l \in Br_{fa.ta}^{FB}} (\max(PTDF_{cb,fa}^t - PTDF_{cb,ta}^t, 0) * f_l^{t,+}) - \sum_{l \in Br_{fa.ta}^{FB}} (\min(PTDF_{cb,fa}^t - PTDF_{cb,ta}^t, 0) * f_l^{t,-}) \le RAM_{cb}^t, \ \forall cb \in CBA, t \in T \tag{31}$$

Eventually, when the patch can no longer reduce adverse flows or CBA is empty, the iterative process is terminated.

2.2. Sub-problems

After the first run of SWMP, the solution contains some PABs and PAMICs (accepted bids with violated the market rules) which can only be identified after the market clearing process because now the price is known. For this reason, the model implements two sub-problems, namely Paradoxically Accepted MIC Sub-Problem (PAMIC-SP) and Paradoxically Accepted Block Sub-Problem (PAB-SP) to eliminate all PABs and PAMICs before moving to PUN-SP, to look for PUN^t. Conversely, there may be some block and MIC orders that are eliminated although they satisfy the acceptance rules (PRBs and PRMICs). Thus, two additional iterative processes are implemented to prevent any PRMICs and PRBs, namely Paradoxically Rejected MIC Sub-Problem (PRMIC-SP) and Paradoxically Rejected Block Sub-Problem (PRB-SP) in order to provide a market solution in which the fairness is enhanced.

The flow-chart of the resulting algorithm is shown in Fig. 1. Here, it should be noted that the designed algorithm involves solving the SWMP iteratively. This, for real-size problems, can be very time consuming due to the presence of a large set of binary variables.

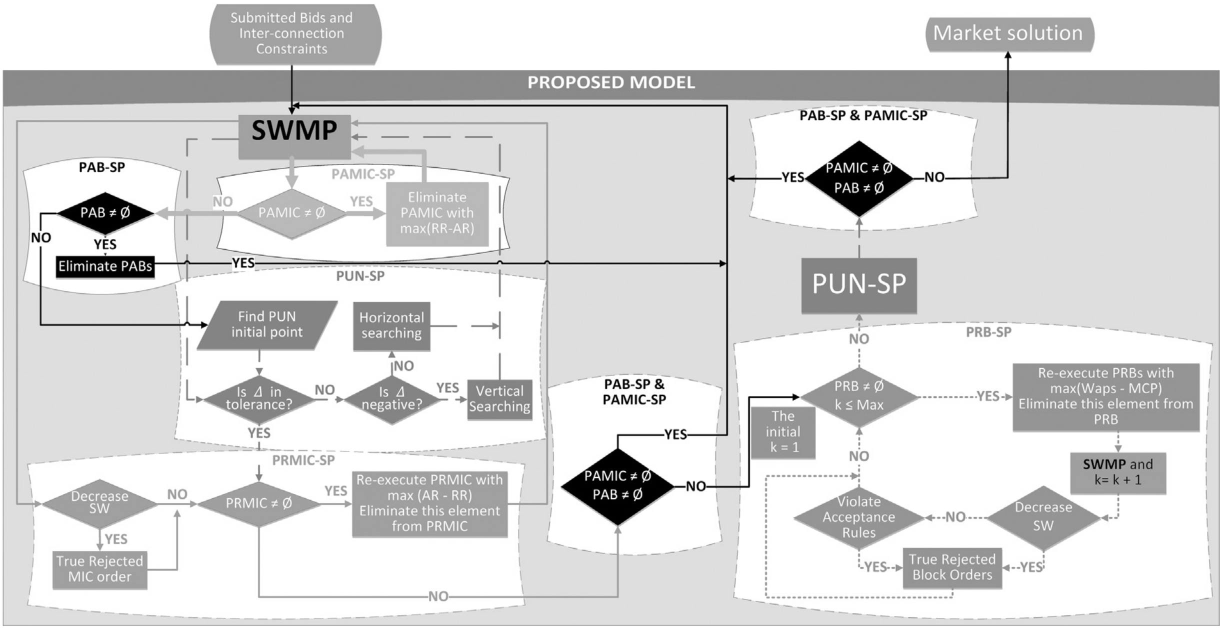

Fig. 1. The flow-chart of the proposed model.

However, since the non-convexity can be caused by binary variables in the vicinity of the marginal offers, the majority of bids will stay either entirely accepted or entirely rejected during the iterations. For this reason, after the first solution of SWMP, all binary variables except the ones representing the link block orders, are fixed to their optimal values and converted to parameters. Latter, the values of these parameters will be decided by the sub-problems according to the satisfaction of the corresponding acceptance rules. Because of the special treatment of link block orders, the binary variables representing the child in the set lb can not be fixed and the value will be decided by the solution of SWMP. Thus, the set of binary variables is dramatically reduced.

2.2.1. PAMIC sub-problem

The binary parameter $Ymic_p$ is modified to decide which MIC order is considered by (13). Thus, before the first run of the SWMP, the clearing conditions for the MIC orders ($RR_p \leq AR_p$) are evaluated using $MCPint_a^t$ in (1) and (2), instead of MCP_a^t. The orders that violate MIC are neglected from the SWMP by setting $Ymic_p$ to 0, while the others are considered ($Ymic_p$ set to 1). Because the PAMIC filter is applied before the first launch of the SWMP, the number of PAMIC orders to be processed by the PAMIC-SP is reduced and, as consequence, the number of PAMIC-SP iterations is reduced.

Then, after the solution of SWMP, RR_p and AR_p are evaluated using MCP_a^t (the Lagrange multiplier of (14)), and $Ymic_p$ is updated. If there are MIC orders that result paradoxically accepted (PAMICs), i.e. $Ymic_p = 1$ and $RR_p > AR_p$, then the one with the maximum violation is deactivated, i.e. $Ymic_p = 0$ for $\max(RR_p - AR_p)$, and the SWMP is run again (the thick continuous line in Fig. 1). Otherwise, the algorithm continues with the PAB-SP module (see next paragraph).

This module is an improved version of the official EUPHEMIA model [8] due to the application of the pre-filtering procedure. Moreover, this approach is straightforward and robust since it avoids the use of user-defined thresholds as in [21,28]. Here, parameters X and Y are used to filter for the PAMICs in a heuristic manner: they can easily introduce PRMICs that will increase the unfairness for MIC orders.

2.2.2. PAB sub-problem

The set of PABs, i.e. PAB_b, is obtained for the regular and flexible hourly block orders as a subset of B_a. Eq. (3) is evaluated using the latest MCP_a^t and Wap_b. Thus, set PAB_b is the set of block orders for which Wap_b is lower (for suppliers) or higher (for demand) than the submitted price of b. Then, all the Y_b in the PAB_b are forced to zero for the next iteration. When the updated PAB_b is empty, the problem continues with the next module, the PUN-SP.

To be noted that the profile block orders can be partially accepted, case in which the acceptance rule is automatically fulfilled, while the exclusive group orders are a particular case of regular block orders and, hence, they are correctly treated by the PAB-SP. Moreover, the linked block orders are treated differently as discussed previously.

2.2.3. PUN search Sub-Problem

The criteria to evaluate if PUN^t is feasible or infeasible is PUN imbalance (Δ^t) because it reflects the difference of payment between two methodologies (MCP_a^t and PUN^t). The tolerance of Δ^t must be small and in between −1€ and 5€ [10]. Therefore, the PUN balance equation is:

$$\Delta^t + \sum_{a \in A} \sum_{ss \in SS_a^{pun}} MCP_a^t * Q_{ss}^t * x_{ss}^t = PUN^t * \sum_{a \in A} \sum_{ss \in SS_a^{pun}} Q_{ss}^t * x_{ss}^t, \quad \forall t \in T \tag{32}$$

As discussed, PUN price (PUN^t) is also a cause of non-convexity as SWMP can find solution with Δ^t out of bounds. Thus, after the PAB-SP stops, the PUN-SP is executed to obtain MCP_a^t that satisfies the PUN balance constraint (32) within the tolerance of Δ^t. In general, the PUN-SP is a process where the demand is iteratively fixed at various levels while reducing the PUN imbalance Δ^t until it gets inside the tolerance. As of today the PUN orders are only hourly step-wise orders, so the solution can be found by performing either a vertical search (constant demand and different PUN^t), or a horizontal search (constant PUN^t, different levels of the accepted demand for the bids with $P_{ss}^t = PUN^t$).

The PUN-SP originally developed in [33] was also adapted in [8]. In these implementations, it is assumed that the initial PUN^t is zero and the accepted demand is maximum; then the accepted demand is gradually decreased and PUN^t increased until Δ^t is inside the tolerance. Obviously, the PUN-SP is quite time-consuming when

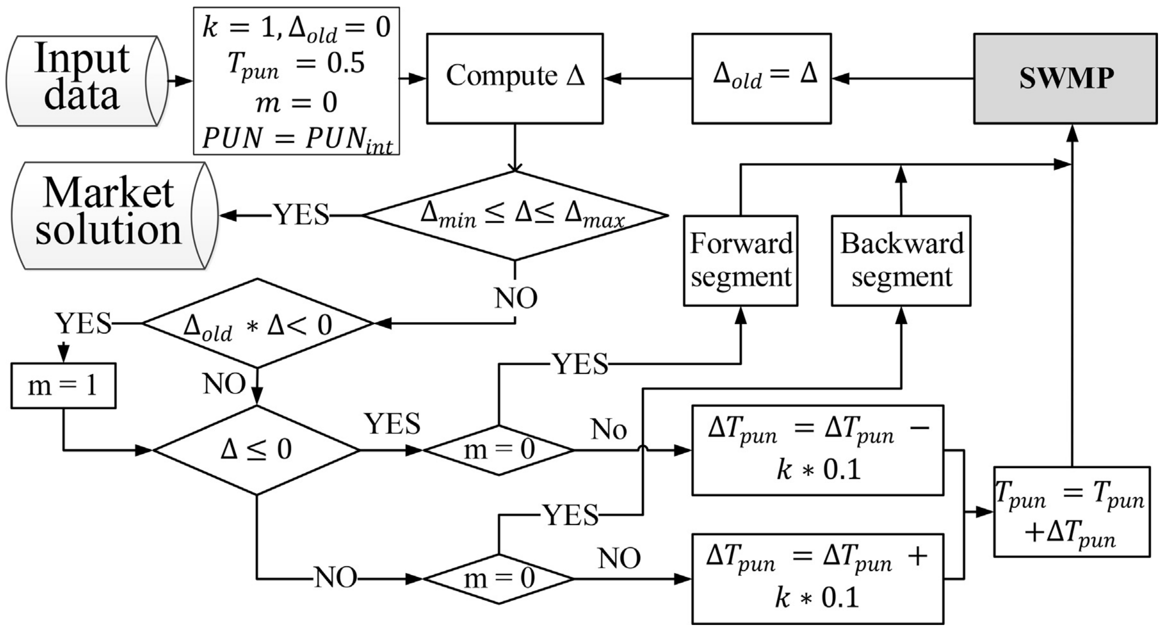

Fig. 2. The flow chart of PUN-SP.

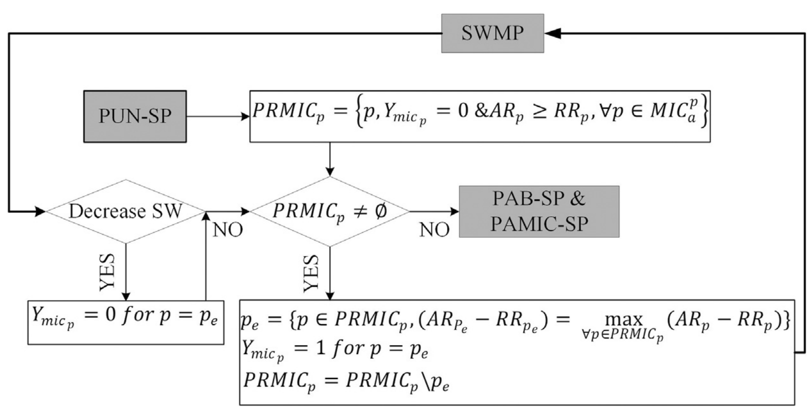

Fig. 3. The flow chart of PRMIC-SP.

the problem size becomes large because of the subsequent calculations. Consequently, the PUN-SP might require a large execution time, above the 10 min requirement specified in public documents [10], which may cause the stop of the PUN-SP to an unsatisfactory solution, where Δ^t is outside the tolerance.

In this article, the authors have developed an improved PUN-SP with the aim to minimize the computation time while satisfying the tolerance of Δ^t. Notably, two are the factors that impact the computation time: (i) the choice of the initial point for PUN^t, i.e. $PUNint^t$, and (ii) the horizontal search strategy. Thus, $PUNint^t$ is calculated based on the available market solution according to (33):

$$PUNint^t = \frac{\sum_{a \in A} \sum_{ss \in SS_a^{pun}} MCP_a^t * Q_{ss}^t * x_{ss}^t}{\sum_{a \in A} \sum_{ss \in SS_a^{pun}} Q_{ss}^t * x_{ss}^t}, \quad \forall t \in T \tag{33}$$

Then, a flexible horizontal search has been defined where the search direction and the incremental step of T_{pun}^t are adaptable to minimize the number of iterations. Here, parameter T_{pun}^t defines the total accepted demand volume of the horizontal segment on which the search is performed. To impose this quantity, Eq. (12) is applied for the bids forming the horizontal search, i.e. for the bids forming the set PUN_{eq}^t, dynamically defined by the algorithm. The x_{ss}^t variables for the bids outside PUN_{eq}^t are converted into constant parameters according to the value of $PUNint^t$: if the submitted price is higher than $PUNint^t$, the bid is fully accepted, otherwise the bid is fully rejected. The flow-chart of PUN-SP is illustrated in Fig. 2 and the main steps are:

1. *Initial point*: The initializations are: PUN_{eq}^t is empty, $PUN^t = PUNint^t$ and Δ^t is computed. If Δ^t is outside the tolerance, the *Vertical search* is initialized: (i) if Δ^t is positive, select the backward segment on the demand curve, while (ii) if Δ^t is negative, select the forward segment on the demand curve. If Δ^t is inside the tolerance, exit the PUN-SP.
2. *Vertical search:* Fix the demand to the selected vertical segment, then solve SWMP and find the PUN^t that minimizes the imbalance Δ^t in (32). If the new Δ^t is outside the tolerance then: (i) if Δ^t changed the sign with respect to the previous iteration then perform *Horizontal search*, else (ii) if the new Δ^t is positive then select the backward vertical segment of the demand curve and repeat *Vertical search*, while (iii) if Δ^t is negative then select the forward vertical segment on the demand curve and repeat *Vertical search*. If the new Δ^t is inside the tolerance, then PUN-SP stops.

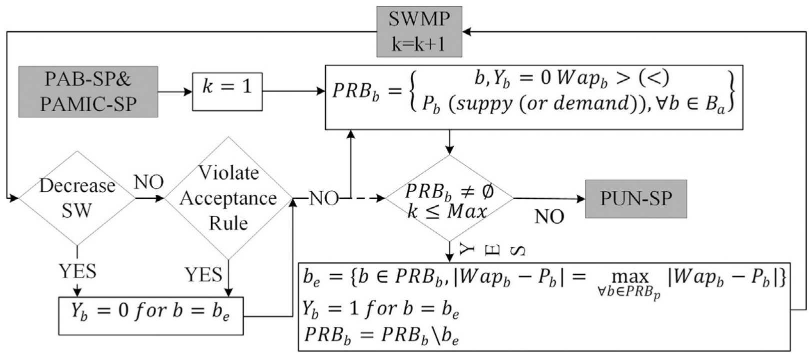

Fig. 4. The flow chart of PRB-SP.

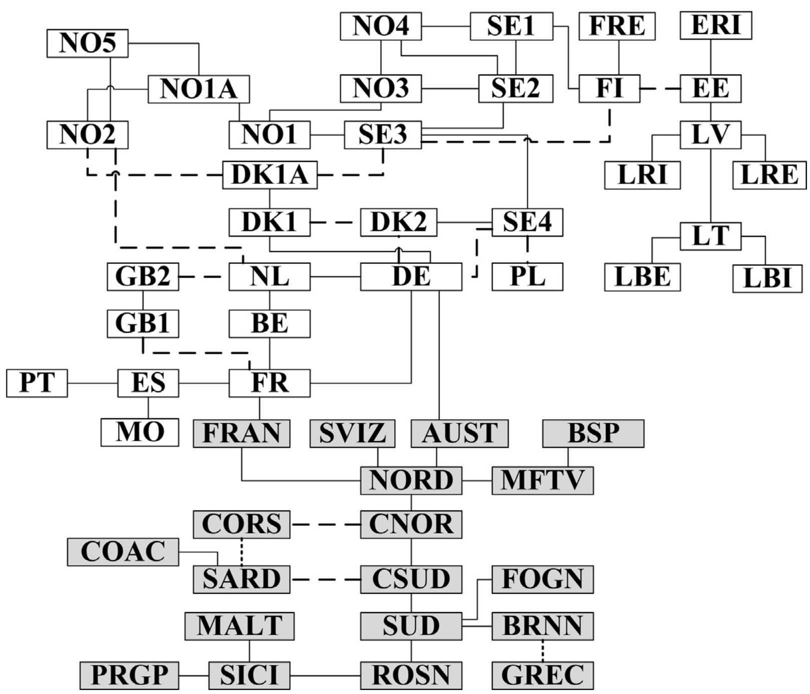

Fig. 5. European system topology.

3. *Preparing for horizontal search:* Parameter T^t_{pun} is set to 0.5, PUN^t is fixed to the price of the considered horizontal segment (P^t_{ss} = PUN^t) and the set PUN^t_{eq} is defined for the sub-orders which have P^t_{ss}=PUN^t. Then, SWMP is solved and the new Δ^t is obtained. If Δ^t is outside the tolerance then: (i) if Δ^t is positive then the incremental step of T^t_{pun}, ΔT^t_{pun} is set to +0.1, else (ii) if Δ^t is negative then ΔT^t_{pun} is set to −0.1. If Δ^t is inside the tolerance, the PUN-SP stops.
4. *Horizontal search:* Update $T^t_{pun} = T^t_{pun} + \Delta T^t_{pun}$, solve SWMP and compute the new Δ^t. If Δ^t is outside the tolerance then (i) if Δ^t does not change the sign with respect to the previous iteration then repeat *Horizontal search*, else change the sign of ΔT^t_{pun} and reduce its value according to a user-defined strategy (adjusting parameter *k* in Fig. 2); repeat *Horizontal search*. If Δ^t is inside the tolerance, the PUN-SP stops.

It should be noted that in Fig. 2, the selection between *Vertical* and *Horizontal search* is made by the parameter *m*, 0 for *Vertical search* and 1 for *Horizontal search*.

The above proposed PUN-SP is run in parallel for all the considered periods. Thus, the rules shown above are applied independently to each period and once per PUN-SP iteration. Hence, the implementation significantly reduces the computation time.

2.2.4. Paradoxically rejected MIC sub-problem

After the execution of PUN-SP, it is now possible, due to the early deactivation of the PAMICs and PABs, to have PRMICs and PRBs. First, the PRMICs are managed by the PRMIC-SP shown in Fig. 3. The designed iterative procedure is:

1. The set of PRMICs, i.e. $PRMIC_p$, is defined for the MICs for which $Ymic_p = 0$ and $AR_p \geq RR_p$;
2. If $PRMIC_p$ is empty then the PRMIC-SP stops and the main algorithm continues. Otherwise, in analogy with the rules defined for PAMIC-SP, the PRMIC with the maximum profit is re-inserted, i.e. $Ymic_p$ is set to 1 for $\max(AR_p - RR_p)$ and set $PRMIC_p$ is correspondingly reduced by one element. The procedure continues to **3**.
3. The SWMP is solved and the impact of the re-inserted PRMIC on the SW is evaluated: (i) if the SW decreased then the re-inserted PRMIC is actually a truly rejected MIC order hence $Ymic_p$ is set to 0; (ii) if the SW increases, or remains constant, then the re-insertion of the PRMIC is correct. The procedure returns to **2**.

Table 1
The structure of the orders in *case study 1*.

Type	Order	Quantity
Single hourly step order	Complex	20,160
	PUN	18,387
	Normal	71,898
Single hourly piecewise order	Normal	4139
Block order	Fill or kill	2061
	Profile block	801
	Flexible	46
	Link Block	176
	Exclusive	207

Table 2
Configurations and computation time.

Config 1	Initialization	*X* and *Y*	Time
1	No	No	60 min
2	Yes	No	52 min
3	No	Yes	55 min

After the execution of the PRMIC, it is now possible, due to the changes in MCP_a^t, to have new PAMICs and/or PABs. If this is the case, then the SWMP, PAMIC-SP, PAB-SP, PUN-SP, and PRMIC-SP are repeated as shown in Fig. 1 and detailed in the previous paragraphs. Otherwise, the algorithm continues with the PRB-SP.

2.2.5. Paradoxically rejected block sub-problem

Finally, the PRB-SP in Fig. 4 is executed by following the procedure:

1. The set of PRBs, i.e. PRB_b is obtained as a subset of B_a. This is the set of block orders for which $Y_b = 0$ and Wap_b is higher (for suppliers) or lower (for demand) than the submitted price of *b*.
2. If PRB_b is empty, then the PRB-SP stops and the main algorithm continues. Otherwise, the PRB for which the distance between the Wap_b and MCP_a^t is maximum, is re-inserted. In other words, Y_b is set to 1 for the block order *b* of PRB_b for which $|Wap_b - MCP_a^t|$ is maximized. The process continue to **3**.
3. The SWMP is solved and the impact of the re-inserted PRB on the SW is evaluated: (i) if the SW decreased then the re-inserted PRB is actually a truly rejected PRB, hence Y_b is set to 0; (ii) if the SW increases, or remains constant, then the acceptance rule is verified again: if the rule is certified then the re-insertion of the PRB is correct, otherwise the re-inserted PRB is actually a truly rejected PRB, hence Y_b is set to 0. The procedure returns to **1** to update the PRB_b set until PRB_b is empty.

After the execution of the PRBs, it is now possible, due to the changes in the MCP_a^t to have Δ^t outside tolerances or PAMICs and/or PABs. If this is the case, the SWMP, PAB-SP, PAMIC-SP and PUN-SP are repeated as shown in Fig. 1. Otherwise, the algorithm will stop and extract the market solution.

It should be noted that the major difference of the proposed PRB-SP and the official EUPHEMIA 9.3 [10] is that the module will generate the list of PRBs continuously (the continuous line in Fig. 4) instead of being created only one time (the dashed line in Fig. 4) at the start of PRB-SP. This adoption helps the module to save more true PRBs than the official one.

3. Test results

The proposed model has been implemented in GAMS environment and CPLEX solver [34] was used to find the solution of the problem. To inquire the robustness and reliability of the proposed model, the authors evaluated two case studies:

Table 3
The summarize of the Config 1.

PAMIC-SP (loops)	PAB-SP (loops)	PUN-SP (loops)	PRMIC-SP (loops)	PRB-SP (loops)	SW (M€)
27	11	24	0	102	6821.759

Table 4
PRB-SP results for Config 1.

PRB-SP	Loops/Max of PRB-SP	True PRBs	SW (M€)	Time (min)
Proposed	102/150	28	6821.659	60
	15/15	9	6821.571	28
Official [10]	100/150	26	6821.472	59
	15/15	8	6817.334	28

- **Case study 1** – *randomly generated bids case*: a realistic market structure of the European DAM containing 117,492 bids, 67 interconnectors, and 53 bidding areas of six PXs (Nordpool, APX, Belpex, EPEXSPOT, OMIE, and GME) has been created. The generated bids include all types of bids and network constraints that can be encounter in the actual European DAM. The aim of this case is to assess the robustness of the proposed model;
- **Case study 2** – *realistic bids case*: this case contains only single hourly step orders obtained directly from the real demand and supply curves of six PXs in Europe such that these curves are correctly emulated. Moreover, the PTDFs and capacity constraints are the real ones obtained from the official websites. This case is used to check the authenticity of congestion allocation in the model and of the PUN-SP.

All simulations have been carried out on a PC with an Intel(R) Core(TM) i7-4510 CPU @ 2.0GHz processor and 8 GB RAM.

3.1. Case study 1

This case study was developed based on the state of the art of the European DAM. Fig. 5 shows the topology, in which the dashed lines denotes DC interconnectors, while the continuous lines stand for AC interconnectors. The parameters of the ATC and FB models, ramping and the initial value of net position and interconnectors are built artificially, such that to reflect the reality. Likewise the real European DAM, the FB model is applied for France, Germany, Belgium and Netherlands, i.e. for CWE, while the rest of the bidding zones belong to the ATC model. Block orders are generated for CWE and Scandinavia countries, PUN orders for Italy (in CNOR, CSUD, NORD, SARD, SICI, and SUD zones), while complex orders are applied to Spain (ES) and Portugal (PT). In particular, the complex orders belong to 46 producers subjects to MIC, SSC, and LGC (one producer can combine two or three mentioned conditions). The block orders are generated in the same way as in [28,30]. To sum up, a total of 117492 bids for all bidding areas were created, as detailed in Table 1.

Moreover, some specific network constraints are considered for this case study in accordance with the real European DAM. Currently, Nord Pool uses both capacity limits on the individual interconnectors and aggregated limits on sets of interconnectors which represents bottlenecks and have a significant impact on the price calculation [35]. For this reason, lines DK1A-NO2 and DK1A-SE3 form set *ls* on which constraints (24) and (25) are applied. The hourly net position ramping up and down are implemented in the FB region because of the high variation of hourly net position in FB market coupling. Additionally, the model considers that in Italy, the DC interconnector between CNOR and CORS is activated only if DC interconnector between SARD and CSUD is congested [2] by

Table 5
Total energy trading for Config 1.

Supply energy	3,453,847.881 MWh
Demand energy	3,453,182.558 MWh
Losses energy	665.323 MWh

introducing a different value of c_l[1] for the interconnector between CNOR and CORS.

In order to test the robustness of the proposed model in its various aspects, the algorithm was executed with different set-ups in three configurations, as shown in Table 2. The **Config 1** focuses on the operation of all modules, excepts PRMIC-SP. The **Config 2** and **3** reflect different ways to clear PAMICs to test the fairness of MIC orders, because these methodologies can accidentally introduce PRMICs. These settings can be made without the use of $MCPint_a^t$ (**Config 1** and **3**) or with (**Config 2**) as discussed in Section 2.2.1. Moreover, the PAMIC-SP can be executed as proposed in this paper, i.e. without the use of the X and Y parameters proposed in [21,28] (**Config 2**), or using the X and Y parameters (**Config 3**). Finally, it should be noted that $MCPint_a^t$ was obtained from the results of **Config 1** due to the lack of data from previous DAM section. For **Config 2**, the authors analyze the sensitivity to $MCPint_a^t$ to verify the proposed model in case of the deviation of $MCPint_a^t$ within $\pm$ 5%, with respect to the initial one. The sensitivity of parameters X and Y is investigated for **Config 3**, the parameter Y is constant and equal to 3 while parameter X varies from 0.1 to 0.9. In addition, to obtain the minimum Δ^t, all configurations set the minimum value of ΔT_{pun}^t in the **Horizontal search** of PUN-SP equal to 0.0001.

Considering **Config 1** as the base case, Table 3 summarizes the number of times that each module of the proposed model was executed. The proposed model converged and found the feasible solution after 164 executions of the SWMP where the maximum execution time for each iteration is 50 s, the minimum is 19 s, while the average is 22 s (60 min in total). At the final solution, the social welfare is 6821.759 million € and there is no non-intuitive flow in the final result. The final imbalance Δ^t is null for 23 periods (*Vertical search*), and 35 € for one period (*Horizontal search*); even if the non-null PUN imbalance is outside the tolerance for one period, it is not damaging for the market participants as an artificial increase of MCP_a^t by 0.0001 €/MWh will bring the PUN imbalance to null.

In fact, the PRB-SP is the most time consuming module as it represented the largest loop in Table 3. Therefore, the proposed model in **Config 1** is tested with two different PRB-SPs, the proposed PRB-SP and the Official EUPHEMIA PRB-SP [8] with or without the time computation limit. Unfortunately, the EUPHEMIA PRB-SP module as with the entire EUPHEMIA modules is not publicity available, only a general description is [8]. As consequence, the proposed PRB-SP was adapted to represent, as much as possible the main characteristics of EUPHEMIA. This adaptation is further known as the *Official model* and here the set PRB-SP is created only one time at the beginning of the module.

Table 4 shows the comparison between them when **Config 1** is run. Especially, the last column of Table 4 reports the computation time for each setting. Without limiting the number of PRB-SP iterations, both approaches should converge to the same, feasible solution (no PABs, PAMICs, PRBs and PRMICs) for which the SW is 6821.759 million €. In fact, the model with the proposed PRB-SP found this solution in 102 iterations and processed 28 true PRBs, while the model with the Official PRB-SP converged in 100 iterations, processed 26 true PRBs but found a slightly worse SW (187 k€). Apparently, the model with the Official PRB-SP is better as it converges in slightly less iterations. However, the total number of iterations of both approaches is too high for a real time implementation in terms of computation time (both approaches would require more than 10 min to converge). This is normal considering the dimensions of the problem in term of number of the variables and constraints.

It should be reminded that to find the solution of the proposed model, a general purpose solver (CPLEX) was used. Meanwhile, the solver for the EUPHEMIA model is a highly optimized in-house solution. Therefore, it is very difficult to evaluate the performance of module directly from the computation time point of view, but it is possible to use instead number of iterations limit. Therefore, the parameter *Max* in PRB-SP is set at 15 as recommended in [10]. In this case, both approaches converged in the same time (28 min), but the proposed approach shows better performances (see Table 4): the final SW is 6821.571 million € and 9 true PRBs are processed, while the SW for the model with the Official PRB-SP [10] is 6817.334 million € and 8 true PRBs are processed. Therefore, with the proposed PRB-SP, a SW much closer to the ideal solution is found and more true PRBs are identified with respect to the Official PRB-SP.

The final solution of **Config 1** contains two block orders for which the acceptance rule of block order (3) is violated. Actually, they are link block orders and are parents of 2 children block orders: the surplus of children compensates for the loss of SW by the parent. Moreover, Table 5 shows the total energy trading obtained for the considered day, including the produced, consumed and lost energy. As one can see, it is balanced and so the final solution checks.

In **Config 1**, after 27 loops of PAMIC-SP, there are 19 over 46 MIC orders with $AR_p \geq RR_p$ and all are accepted, and there is no PRMICs.

Now, the proposed model is run with **Config 2** and **3**, and the *Max* parameter of PRB-SP is set at 150.

In **Config 2**, the $MCPint_a^t$ is obtained from the MCP_a^t of **Config 1**. Here, the sensitivity analysis of $MCPint_a^t$ is investigated by using $MCPint_a^t * k$ as input with $k \in [0.95, 1.05]$. Two situations can occur: (i) the model finds 27 PAMICs ($AR_p < RR_p$), and 19 MIC orders have $AR_p \geq RR_p$ if the value of k is greater than 1 (RR_p is constant, while AR_p increases because $MCPint_a^t$ increases), thus the PRMIC-SP does not activate; (ii) the model finds 30 PAMICs ($AR_p < RR_p$), and only 16 MICs have $AR_p \geq RR_p$ if the value of k is less than 1; hence, the model activates the PRMIC-SP which is executed 3 times to guarantee that the market solution contains 19 MICs as **Config 1**.

In **Config 3**, the sensitivity analysis of parameter X between 0.1 and 0.9 with the incremental step of 0.1 is executed. Two cases can happen: (i) if the value of parameter X is smaller than 0.3, PAMIC-SP finds 30 PAMICs and 16 MIC orders have $AR_p \geq RR_p$; (ii) the PAMIC-SP finds 34 PAMICs, and 12 MIC orders have $AR_p \geq RR_p$. Thus, in both cases, the model activates the PAMIC-SP, which is executed 3 times for case (i) and 7 times for case (ii), and the minimum number of PAMIC-SP loops equas to the value of parameter Y (in this configuration, Y equals to 3).

It is clear that the PRMIC-SP guaranteed the fairness of MIC orders in all cases, therefore the social welfare of **Config 2** and **3** are always similar to the **Config 1** (6821.759 million €). Moreover, the computation time of all **Config** is shown in the last column of Table 2. It should be noted that in **Config 2** and **3**, it only shows the total convergence time.

Eventually, according to Table 2 and 4, it can be concluded that the proposed model work more efficient than the *Official model* in term of computation time and social welfare. The best performance of the proposed model is **Config 2**, where the model takes the least time to converge and obtain the highest social welfare.

3.2. Case study 2

In this case study, an instance (day) of the actual European DAM is simulated, in order to test the model in the real-life situation.

[1] c_l equal to 0.001 for all interconnectors which are in the FB region and DC interconnectors. Especially, c_l of SARD-CSUD equals to 0.002.

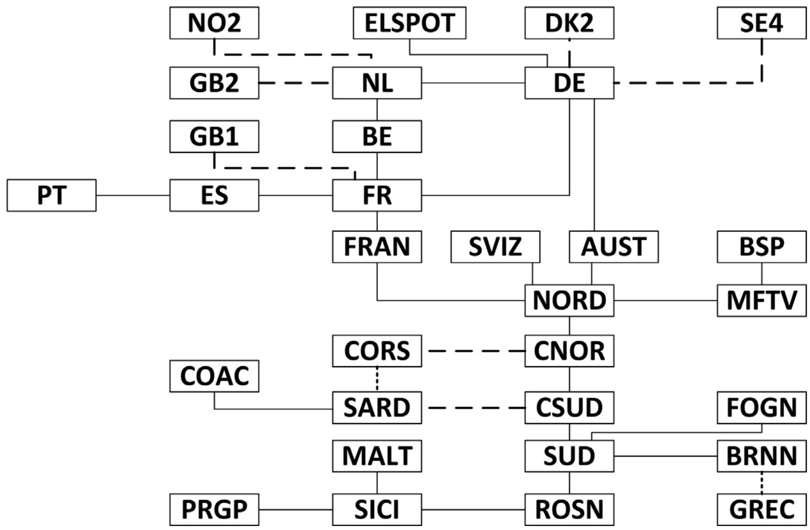

Fig. 6. The European simplify topology.

Table 6
The report of the absolute value of the relative variation of simulated and realistic electricity market price in percentage in CWE and Europe on 9th, March 2016.

Test case		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Max	CWE Sim	0.76	6.64	0.19	0	0.35	1.14	0	0.91	1.15	0.43	0.70	1.01	2.55	1.44	0.04	0.82	0	0	0.57	0.33	3.49	0.54	0.04	0.30
	Europe Sim	2.10	4.32	2.05	1.61	0.84	1.09	0.89	2.56	21.82	4.27	4.43	2.60	4.40	3.15	2.64	2.77	0.78	4.55	0.53	2.61	2.01	2.41	7.05	3.89
Min	CWE Sim	0.76	0.57	0.19	0	0.35	1.14	0	0.85	0.75	0.32	0.3	0	1.17	1.23	0.04	0	0	0	0.57	0.26	3.49	0.54	0.04	0.15
	Europe Sim	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Average	CWE Sim	0.76	2.87	0.19	0	0.35	1.14	0	0.88	0.92	0.38	0.48	0.50	1.77	1.37	0.04	0.43	0	0	0.57	0.29	3.49	0.54	0.04	0.19
	Europe Sim	0.74	1.54	1.34	0.91	0.40	0.60	0.45	0.32	1.40	0.55	0.45	0.45	0.49	0.54	0.33	0.26	0.09	1.02	0.03	0.27	0.41	0.32	1.20	0.19

Table 7
The days and hours activated Horizontal Search in PUN-SP.

Date	Horizontal search Period [h] (T_{pun}^{t})
10/03/2016	6 h (0.797); 7 h (0.593); 20 h (0.816)
13/03/2016	3 h (0.877); 6 h (0.250)
14/03/2016	20 h (0.02)
15/03/2016	5 h (0.275); 13 h (0.812)

The case study from the publicly available data [36–43] is built. To face with the missing data in most of the PXs, data from supply and demand aggregated curves which are always available is generated. The capacity limits of the ATC model were obtained from ENTSO-E [42], while the Flow-Based model parameters (RAM and PTDFs) from [43]. The topology of European DAM in Fig. 6 was simplified from Fig. 5, based on the available public data. The authors selected 9th, March 2016 to build the case study. Because of the missing data, the input data contains only hourly stepwise and piecewise orders, and all network constraints of Fig. 6.

First, the generated supply and demand bids are validated by running the SWMP for each bidding area independently, considering the power flows on the inter-connector constant injections and neglecting the corresponding equations. The obtained solution is compared with the real market results in terms of clearing prices: the errors are very small (below 1%) proving that the bids were generated correctly.

Next, the interconnectors between these bidding areas are considered and the realistic European DAM is simulated. The main objective of this simulation is to test the robustness and accuracy of the proposed model regarding: (i) the adopted FB methodology and hybrid model, (ii) PUN price, and (iii) the clearing prices obtained across Europe. For this reason, two simulations were run: (i) only the FB area market (*CWE Sim*), and (ii) the entire data (*Europe Sim*). Table 6 shows the results in term of the absolute value of the relative variation of MCP_a^t between simulation and the reality in percentage for the maximum, minimum, and average value for 24 periods in the two simulations. The result shows that for *CWE Sim*, the maximum value belongs to France in period 2 with 6.64%, while the average error is 0.72%. Definitely, the above result guarantees that the data has enough reliability to simulate the European DAM for 9th, March 2016. The results obtained for *Europe Sim* are very close to the real ones showing the reliability and robustness of the model: generally, the average error is bellow 1% with the maximum being equal to 1.5%. Moreover, in 315 over 960 cases, the error is null. Also, the maximum error is well contained, around few %. However, in one case i.e. for hour 9 and area ELSPOT the difference between the real and simulated clearing prices is high, i.e. 22%. This is because the simplified topology adopted here prevented the simulations to catch a congestion between ELSPOT and CWE areas: the real topology is meshed (Fig. 5) while the simplified one is radial (Fig. 6). In particular, for *Europe Sim* the prices in the ELSPOT area and in the DE area (the only area with whom ELSPOT is connected in the simplified topology) are equal, i.e. 31 €/MWh, while in reality they are different due to a congestion, i.e. 39.65 €/MWh for ELSPOT and 29.91 €/MWh for DE. However, in all the other cases the differences between reality and simulation are very small.

Moreover, the authors evaluated the proposed PUN-SP with the realistic data for one week in [36]. The PUN-SP caught exactly PUN^t and T_{pun}^t within 1 min in all cases. Table 7 represents the hours of days when *Horizontal search* in the PUN-SP was activated.

4. Conclusion

In this paper, an European DAM model has been successfully developed. The model has been formulated as an MIQCP model, implemented in GAMS, and CPLEX solver was used to find the solution. A heuristic iterative algorithm is implemented to mitigate the non-convexity of the regular blocks, MIC and PUN orders. The proposed model has significant contributions to the correlation of the SWMP, PAB-SP, PAMIC-SP, PUN-SP, PRMIC-SP and PRB-SP module, in which MIC and PUN orders are cleared flexibly. The proposed model proved on par with the official model when computation time was not limited, while it provided a better and much fair solution when the computation time was limited by limiting the number of iterations. Thus, the proposed model is suited for real-life application.

The authors used two case studies to verify the robustness and reliability of the proposed model. The result showed that the model works well with large numerical data, of the same size as the real one. Additionally, the simulation of the current DAM price can confirm the security of congestion allocation mechanism.

For further research, the authors will put attention on the proposal of non-uniform price in the DAM and the coupling of Intra-day markets in Europe.

Appendix A. The functions of orders

$$c_{ss}^t = x_{ss}^t * Q_{ss}^t * P_{ss}^t;\ \forall ss \in SS_a, t \in T \tag{A.1}$$

$$c_{sp}^t = Q_{sp}^t * x_{sp}^t * (P0_{sp}^t + \frac{P1_{sp}^t - P0_{sp}^t}{2} * x_{sp}^t);\quad \forall sp \in SP_a, t \in T \tag{A.2}$$

$$c_b = \sum_{pb \in b} x_b * Q_{pb} * P_b * (\underline{T_{pb}^t} - \bar{T}_{pb}^t + 1);\ \forall b \in B_a \tag{A.3}$$

$$c_{fh}^t = x_{fh}^t * Q_{fh}^t * P_{fh}^t;\ \forall fh \in FH_a, t \in T \tag{A.4}$$

Appendix B. The functions of power flow

$$f_l^{t,in} = f_l^{t,+} * (1 - loss_l) + f_l^{t,-},\ \forall t \in T, l \in Br_{fa.a}^{DC} \tag{B.1}$$

$$f_l^{t,out} = f_l^{t,-} * (1 - loss_l) + f_l^{t,+},\ \forall t \in T, l \in Br_{a.ta}^{DC} \tag{B.2}$$

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